EMI
QUICK REFERENCE

The complete loan-payment manual for Indians

SaralArthaShastra · By Ajit Kumar

Every formula, every example, on one page

The EMI Formula

EMI = P × r × (1+r)^n / ((1+r)^n – 1)

P	Principal — Loan amount (e.g., ■50,00,000)
r	Monthly interest rate = Annual Rate / 12 / 100
n	Total months = Years × 12
EMI	Equated Monthly Instalment (your monthly payment)

Worked Example

Loan: ■50,00,000 · **Rate:** 8.5% per year · **Tenure:** 20 years

Step 1: Convert rate to monthly → r = 8.5 / 12 / 100 = 0.00708
Step 2: Convert tenure to months → n = 20 × 12 = 240
Step 3: Apply formula:
EMI = 5000000 × 0.00708 × (1.00708)^240 / ((1.00708)^240 – 1)
EMI = 5000000 × 0.00708 × 5.45 / 4.45
EMI = ■43,391 per month

EMI per ■1 Lakh (Quick Reference)

Multiply these by your loan amount in lakhs to get your EMI:

Rate ↓	5 yrs	10 yrs	15 yrs	20 yrs	25 yrs	30 yrs
7%	■1,980	■1,161	■899	■775	■707	■665
7.5%	■2,003	■1,187	■927	■806	■739	■699
8%	■2,028	■1,213	■956	■836	■772	■734
8.5%	■2,052	■1,240	■985	■868	■805	■768
9%	■2,076	■1,267	■1,014	■900	■839	■805
9.5%	■2,101	■1,295	■1,044	■932	■874	■841
10%	■2,125	■1,322	■1,075	■965	■909	■878
10.5%	■2,149	■1,351	■1,106	■998	■944	■915
11%	■2,174	■1,379	■1,138	■1,032	■980	■952

Example: ■50 lakh loan at 8.5% for 20 years → 868 × 50 = ■43,400/month (matches our calculation)

How EMI Splits Between Principal & Interest

In early years, MOST of your EMI goes to interest. As principal reduces, more goes to principal. This is why prepayment in early years saves you the most.

Year	EMI	Principal Part	Interest Part	Outstanding
Year 1	■43,391	■7,055	■36,336	■49,15,463
Year 5	■43,391	■9,876	■33,515	■46,53,278
Year 10	■43,391	■14,890	■28,501	■39,72,168
Year 15	■43,391	■22,422	■20,969	■28,52,102
Year 20	■43,391	■33,786	■9,605	■0

The Prepayment Strategy

On a ■50 lakh / 8.5% / 20-year loan, paying just **1 extra EMI per year** (out of bonus or savings) saves you:

Strategy	Total Interest	Years to Close	Savings
No prepayment	■54,13,898	20.0	■0
1 extra EMI/year	■40,87,245	17.5	■13,26,653
2 extra EMI/year	■32,89,521	15.6	■21,24,377
10% of original principal once at year 5	■46,42,801	17.8	■7,71,097

Tax Benefits on Home Loan

Section 24(b): Up to ■2,00,000 deduction on interest paid (self-occupied property)

Section 80C: Up to ■1,50,000 deduction on principal repayment

Section 80EE / 80EEA: Additional ■50,000-1,50,000 for first-time buyers (conditions apply)

Total possible: Up to ■4,00,000+ in deductions per year

Use the live calculator at saralarthashastra.com/tools.html

Punch in your exact numbers — see EMI, total interest, and prepayment savings.